

Answer all questions.

1. Read the information about obesity.

Fact file on Obesity

- In 2015, one in five adults and many children in the UK showed signs of being overweight or obese.
- Obesity makes it more likely that a person will develop medical conditions such as arthritis and type 2 diabetes as well as life-threatening diseases such as heart disease and cancer.
- There are more obese men than obese women but high blood pressure and type 2 diabetes are four times more common in obese women than in obese men.
- A diet low in saturated fat combined with regular exercise can be an effective way to reduce or avoid obesity.

(a) (i) Use the information to complete the table. Write 'true' or 'false' for each statement. [3]

Statement	True or False
In 2015, 20% of the population of the UK was obese.	
Obesity increases the risk of serious diseases.	
Obese women are more likely than obese men to have cells in their bodies which do not respond to insulin.	
A healthy lifestyle will reduce obesity.	

(ii) Complete the following sentence by selecting your answer from the words below. [1]

nerves blood bile

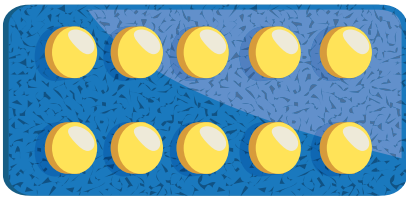
Hormones such as insulin are chemical messengers carried in the



(b) In 2010 a drug, Sibutramine, was developed to help treat obesity.

Ten patients in a hospital tested the drug for a month and doctors observed that it resulted in considerable weight loss.

Medical researchers then carried out a double-blind clinical trial, where 600 patients took the drug for two years. Patients in group **A** received Sibutramine tablets and patients in group **B** received a placebo.



Sibutramine tablets

(i) State **two** differences between the trial carried out by the medical researchers and the test carried out in the hospital. [2]

.....

.....

.....

.....

(ii) State the meaning of the term *placebo*. [1]

.....

.....

(iii) Explain why a placebo is necessary in trials of new drugs. [1]

.....

.....

3400U201
03



- (iv) Choose the letter (**A-D**), which gives the meaning of the term double-blind trial. [1]
- A** Only the doctor knows whether the patient is in group A or group B
 - B** The doctor and the patient both know whether the patient is in group A or group B.
 - C** Neither the doctor nor the patient know whether the patient is in group A or group B.
 - D** Patients can choose whether they wish to be in group A or group B.

Answer

- (c) Before new medicines are tested on human patients they are usually tested on animals in laboratories. It is now possible to test some medicines using computer programs which can model the human body or use tissue cultures from human cells in Petri dishes.

State **one** advantage of using computer programs or tissue cultures rather than animals for testing medicines. [1]

.....

10

